CSCI 5541 NLP HW2

Cheng-Fu Tseng `tseng097@umn.edu`

1 Introduction

In this assignment, we compare different models, which are single-layer perceptron(SLP) and two-layer multi-layer perceptron (MLP). And MLP with different hidden dimension size (100,500). Then test and train on the tweet_eval dataset on the huggingface. Finally, conduct the error analysis.

2 Methodology

2.1 Tokenizer

First, I uses regular expressions to extract all letters and numbers and convert them to lowercase, thereby breaking down a sentence into a list of words without punctuation. And compare with method which utilize the `o200k_base` tokenizer from *tiktoken*(given in hw1) to convert raw text into sequences of tokens.

2.2 Model Details

This model ask us compare model between single-layer perceptron(SLP) and two-layer multi-layer perceptron (MLP). In the both models, the input is a vector which finds the 64-dimensional vector of each word, and then calculates the average value. And the output is a vector which dim is 3 (represent the emotion: Negative, Neutral and Positive). Finally, the MLP have an extra layer which have 100 or 500 features.

3 Experimental Results

3.1 Hyperparameter

Before show the experimental results I consider that I must disclose the hyperparameters I use in the experiment. The most important is different of the hidden layer size(100, 500). After that, I set Learning Rate: 2.0; EPOCHS: 10; BATCH SIZE: 64; Embedding Dimension: 64.

3.2 Accuracy

First, followings are the accuracy in train and test data. As said before, I utilize regular expressions, thereby breaking down a sentence into a list of words without punctuation.

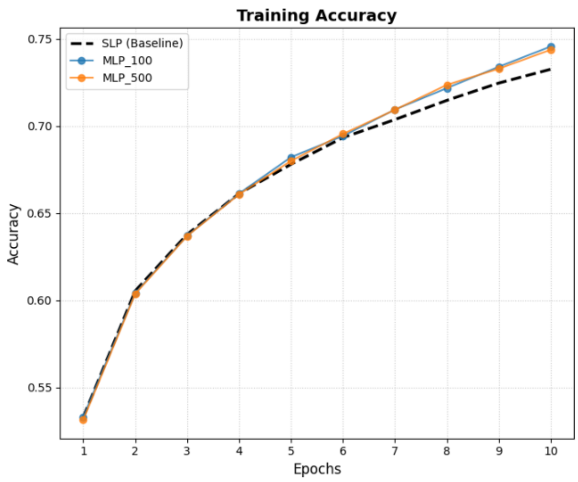

Figure 1: train accuracy

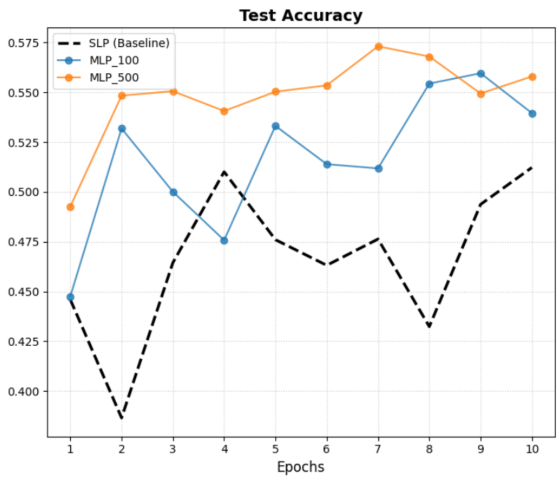

Figure 2: test accuracy

As the result shows above, we can see the accuracy are stably increase in the train data but very unstable in the test data. There seems have overfit in this situation. But still the model with hidden layer has outperformed with the baseline

Second, I choose different tokenizer method the `o200k_base` tokenizer from *tiktoken*. The result is following.

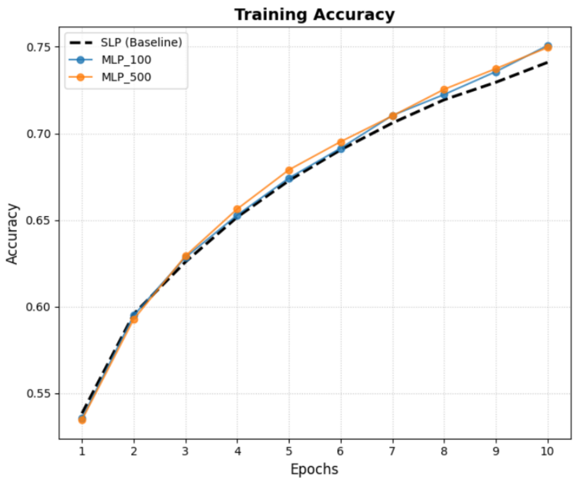

Figure 3: train accuracy with tokenizer

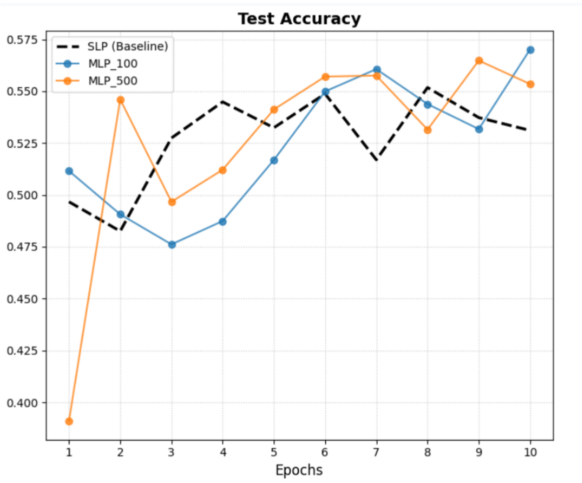

Figure 4: test accuracy with tokenizer

When I use differnet tokenizer method. The path of the train accuracy basic same as using oringinal model. However, the test accuracy seems more robust here. And in this case, it hard to say which model is better.

3.3 Error Analysis

This part is the error analysis compare with the baseline tokenizer method and using 500 hidden MLP. Perform error analysis, assignment ask us provide at least 5 incorrectly classified examples. The following are:

Example 1:
Text: user user that s coming but i think the victims are going to be medicaid <unk>
True Label: Neutral
Predicted Label: Positive

Example 2:
Text: i think i may be finally in with the in crowd <unk> <unk> user
True Label: Positive
Predicted Label: Neutral

Example 3:
Text: user wow first hugo <unk> and now <unk> castro danny glover michael moore oliver stone and sean penn are running out of heroes
True Label: Negative
Predicted Label: Positive

Example 4:
Text: twitter s <unk> shows heartfelt gratitude to potus
True Label: Positive
Predicted Label: Neutral

Example 5:
Text: all <unk> and fracking all around australia is to cease and mining entities held accountable sign the petition
True Label: Neutral
Predicted Label: Positive

The error analysis reveals that the model primarily struggles with context and semantic problems. For instance, Example 1 is mask at critical word ; Example 2 use idiom "in with the in crowd"; Example 3 is using sarcasm; Example 4 obscured critical sentiment indicators; Example 5 I think it misclassifies because of it ambiguity of the action.

4 Summary

In this report, I evaluated various models trained on the dataset and tracked their accuracy across epochs. I also experimented with different tokenization methods. Roughly speaking, the MLP with 500 hidden units performed slightly better than the other configurations. However, I consider that the overfitting seems happened and the results remain unstable, suggesting potential sensitivity to hyperparameters or data variance.

5 References

Dataset: TweetEval (Sentiment Analysis)
https://huggingface.co/datasets/cardiffnlp/tweet_eval/viewer/sentiment